

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL GCSE CHEMISTRY

Paper 2

Monday 3 June 2019

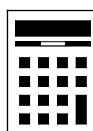
07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

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0 1

This question is about Group 1 metals and their reactions.

0 1 . 1

An atom of lithium can be represented as ${}^7_3\text{Li}$.

How many electrons are in an atom of lithium?

Tick (✓) **one** box.

[1 mark]

1

☐

3

☐

4

☐

7

☐

0 1 . 2

Flame tests can be used to identify metal ions.

Describe how to do a flame test on a solution containing metal ions.

[2 marks]



01.3

Flame tests can be used to identify lithium and sodium ions.

Draw **one** line from each metal ion to the colour of the flame.**[2 marks]****Metal ion****Colour of the flame**

Lithium

Sodium

Blue

Crimson

Green

Lilac

Yellow

Question 1 continues on the next page**Turn over ►**

0 1 . 4 Table 1 gives some data about Group 1 metals.

Table 1

Group 1 metals	Melting point in °C	Density in g/cm ³
Lithium	181	0.53
Sodium	98	0.97
Potassium	63	0.86
Rubidium	39	1.53
Caesium	29	1.88

Describe the **two** trends shown in **Table 1** as you go down Group 1.

[2 marks]

1 _____

2 _____

0 1 . 5 Calculate the volume, in cm³, of a piece of potassium with a mass of 0.43 g.

Use the equation below and **Table 1**.

$$\text{Volume (cm}^3\text{)} = \frac{\text{mass (g)}}{\text{density (g/cm}^3\text{)}}$$

[2 marks]

Volume = _____ cm³



0 1 . 6

Sodium reacts with water to produce sodium hydroxide and a flammable gas.

Write a word equation for the reaction.

[2 marks]

0 1 . 7

Sodium hydroxide is a strong alkali.

What is the pH of sodium hydroxide solution?

Tick (✓) **one** box.

[1 mark]

1

☐

7

☐

8

☐

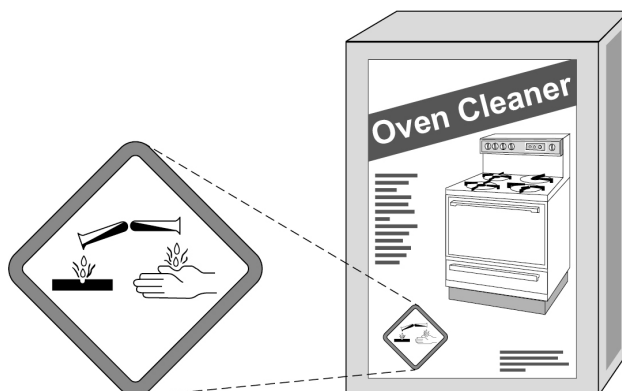
14

☐

0 1 . 8

Oven cleaner contains sodium hydroxide.

Figure 1

The hazard symbol in **Figure 1** shows that the oven cleaner is corrosive.Suggest **one** way to minimise the risk when using oven cleaner.

[1 mark]



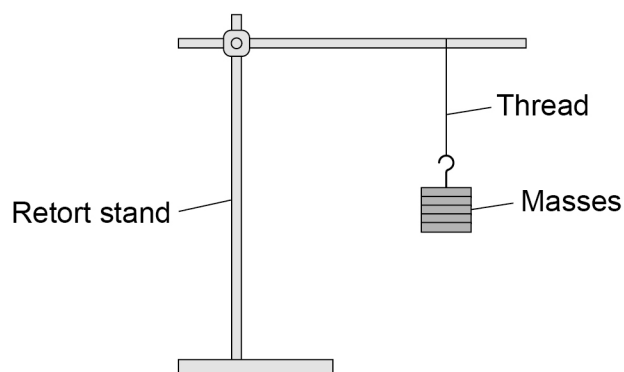
0 2

A student investigated the strengths of different types of thread.

The student used cotton, nylon, polyester and wool threads.

Figure 2 shows the apparatus used.

Figure 2



This is the method used:

- 1 Set up the apparatus shown in **Figure 2** using cotton thread.
- 2 Add 5 g masses until the thread breaks.
- 3 Record the total mass added.
- 4 Repeat steps 1–3 with cotton thread three more times.
- 5 Repeat steps 1–4 for nylon, polyester and wool threads.

Table 2 gives the results of the investigation.

Table 2

Type of thread	Mass needed to break thread in g			
	Attempt 1	Attempt 2	Attempt 3	Attempt 4
Cotton	25	25	30	30
Nylon	45	45	5	45
Polyester	30	25	30	25
Wool	15	15	10	15



0 2 . 1 What were the dependent and independent variables in this experiment?

Draw **one** line from each type of variable to the description of the variable.

[2 marks]

Type of variable	Description of the variable
	Length of thread
	Mass needed to break thread
Dependent variable	Mass of thread
Independent variable	Thickness of thread
	Type of thread used

0 2 . 2 There is an anomalous result in **Table 2**.

Which is the anomalous result?

[1 mark]

Type of thread _____ Attempt _____

0 2 . 3 Suggest a reason for the anomalous result.

[1 mark]

0 2 . 4 Suggest **one** way of improving the accuracy of the results of the experiment.

[1 mark]

Question 2 continues on the next page

Turn over ►



0 2 . 5

Polymer threads made from crude oil are not biodegradable.

Give **one** problem caused by waste polymers.

[1 mark]

0 2 . 6

Nylon is a thermosoftening polymer.

Explain why thermosoftening polymers soften when heated.

[2 marks]

8

Turn over for the next question

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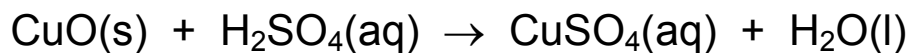
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IB/M/Jun19/9202/2

The student then makes copper sulfate crystals by reacting copper oxide with dilute sulfuric acid.



This is the method used:

- 1 Add 100 cm³ of dilute sulfuric acid to a beaker.
- 2 Warm gently.
- 3 Add excess copper oxide.
- 4 Filter the mixture.
- 5 Collect the filtrate in an evaporating basin.
- 6 Heat the solution to evaporate some of the water.
- 7 Leave to cool.

0 3 . 4 Why is excess copper oxide added in step 3?

[1 mark]

0 3 . 5 Why is the mixture filtered in step 4?

[1 mark]

Question 3 continues on the next page

Turn over ►

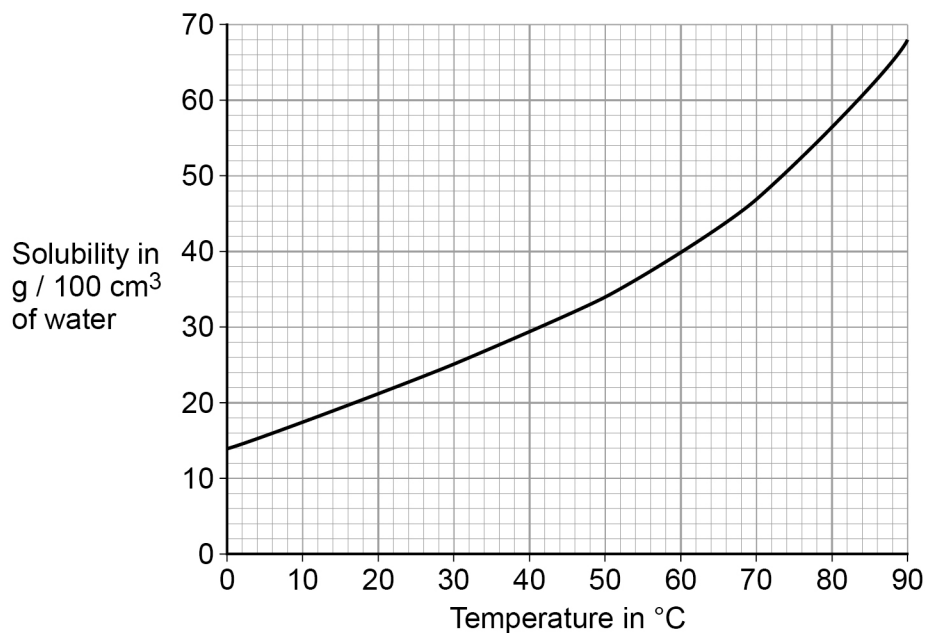


A definition of the solubility of a substance is

‘The maximum mass, in grams, of a substance that will dissolve in 100 cm³ of water.’

Figure 3 shows the effect of changing temperature on the solubility of copper sulfate.

Figure 3



0 3 . 6

Explain why crystals form in step 7 of the method.
Use **Figure 3**.

[2 marks]



03.7

Calculate the minimum volume of water needed to dissolve 16.0 g of copper sulfate at 40 °C.

Use data from **Figure 3**.

[3 marks]

Volume = _____ cm³

Turn over for the next question

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11

Turn over ►



0	4
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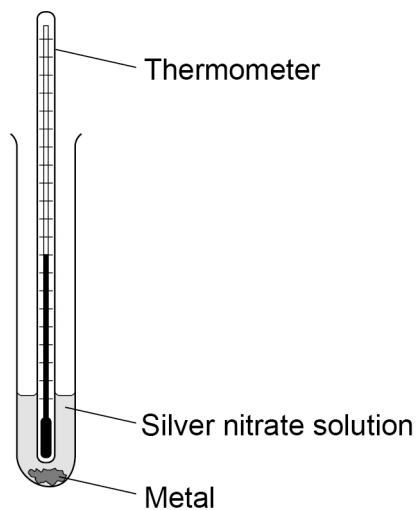
A student investigated the displacement reactions of metals.

Displacement reactions are exothermic.

The student added different metals to silver nitrate solution and measured the increase in temperature.

Figure 4 shows the apparatus used.

Figure 4



0	4
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1

Give **three** variables that should be controlled in this investigation.

[3 marks]

1 _____

2 _____

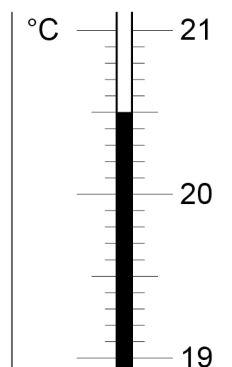
3 _____



0 4 . 2 Zinc was added to silver nitrate solution.

Figure 5 shows the thermometer when the temperature was at its highest.

Figure 5



Complete **Table 3**.

Use **Figure 5**.

[2 marks]

Table 3

Highest temperature in °C	
Starting temperature in °C	
Temperature increase in °C	4.5

Question 4 continues on the next page

Turn over ►



Table 4 shows the student's results for four other metals added to silver nitrate solution.

Table 4

Name of metal	Temperature increase in °C
Copper	1.0
Gold	0.0
Iron	3.0
Magnesium	9.5

0 4 . 3 The student drew a bar chart to represent the results in **Table 4**.

What is the reason for drawing a bar chart rather than a graph?
Tick (✓) **one** box.

[1 mark]

Name of metal is a categoric variable.

☐

Name of metal is a continuous variable.

☐

Temperature increase is a categoric variable.

☐

Temperature increase is a continuous variable.

☐

0 4 . 4 The student repeated the experiment using an unknown metal, **X**.

The reaction resulted in a temperature increase of 2.5 °C.

What can you conclude about the position of **X** in the reactivity series?

Use **Table 4**.

[2 marks]



0 4 . 5

There was no temperature change when gold was added to silver nitrate solution because there was no reaction.

Why was there no reaction?

[1 mark]

0 4 . 6

Suggest **one** reason why the student did not use potassium in this experiment.

[1 mark]

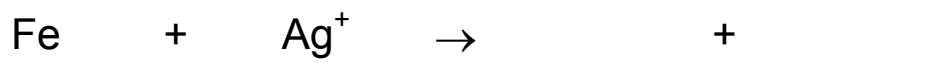
0 4 . 7

The equation for the reaction of iron with silver nitrate solution is

iron + silver nitrate $\rightarrow$ iron(II) nitrate + silver

Complete and balance the ionic equation for the reaction.

[2 marks]



12

Turn over for the next question

Turn over ►



0 5

Ethanol biofuel can be used instead of petrol in cars.

Table 5 gives some information about ethanol biofuel and petrol.**Table 5**

Fuel		Ethanol biofuel	Petrol
Emission rate from car exhaust in grams per kilometre	CO ₂	278	277
	SO ₂	0.02	0.13
Fuel consumption of car in litres per kilometre		0.20	0.14

0 5**1**

Petrol boils over a range of temperatures.

Suggest why.

[1 mark]

0 5**2**

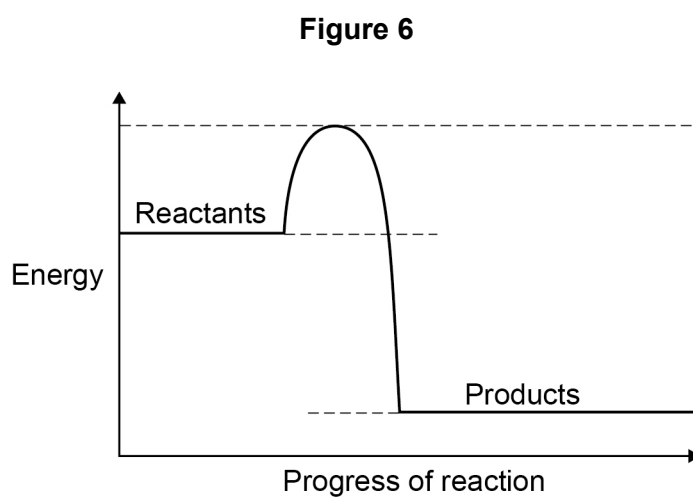
Evaluate using ethanol biofuel compared with petrol as a fuel for cars.

Use **Table 5** and your knowledge and understanding.**[6 marks]**



Extra space

0 5 . 3

Figure 6 shows an energy level diagram for the combustion of ethanol.Label the overall energy change on **Figure 6**.**[1 mark]****Question 5 continues on the next page****Turn over ►**

0 5 . 4

A student did an experiment to measure the amount of energy released by burning ethanol. **Figure 7** shows the apparatus used.

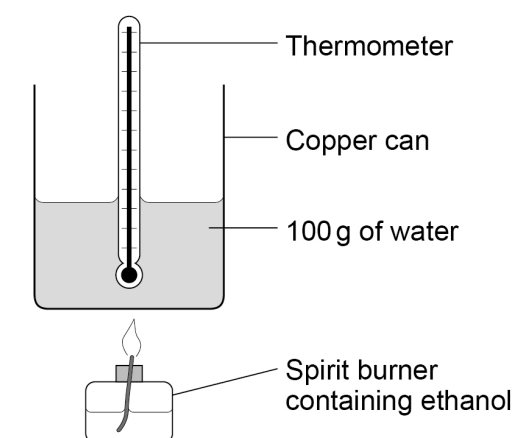
Figure 7

Table 6 shows the student's results.

Table 6

Temperature of water before heating in °C	18.1
Temperature of water after heating in °C	32.5

Calculate the energy released (Q), in joules (J), by ethanol.

Use the equation:

$$Q = mc\Delta T$$

Where:

- Q = energy released in J
- m = mass in g
- c = specific heat capacity of water = $4.2 \text{ J/g } ^\circ\text{C}$
- ΔT = temperature change in $^\circ\text{C}$

[3 marks]

Energy released (Q) = _____ J



0 5 . 5

What is the resolution of the thermometer in **Figure 7**?**[1 mark]**

0 5 . 6

A second student repeated the experiment.

Table 7 shows the mass of the spirit burner before and after the ethanol was burned.**Table 7**

Mass of spirit burner before ethanol was burned in g	75.15
Mass of spirit burner after ethanol was burned in g	74.85

The student calculated the energy released by the ethanol to be 5700 J.

Calculate the energy released in J/g.

[2 marks]

Energy released by the ethanol = _____ J/g

Question 5 continues on the next page**Turn over ►**

0	5	.	7
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Another student did the same experiment and calculated the energy change to be -1013 kJ/mol .

The student compared this value with the data book value of -1367 kJ/mol .

Suggest **one** improvement to the equipment in **Figure 7** that would improve the accuracy of the calculated energy change.

Give a reason for your answer.

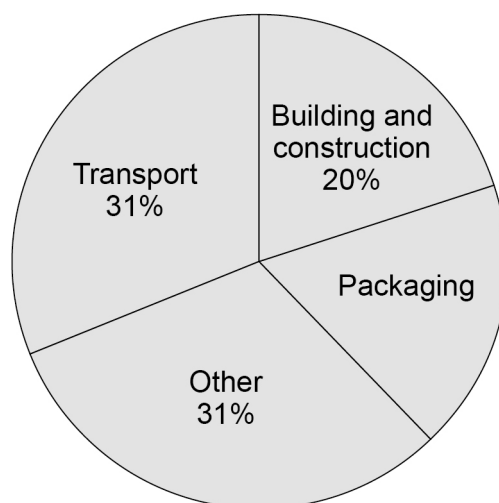
[2 marks]

Improvement _____

Reason _____

16



0 6**Figure 8** shows the common uses of aluminium.**Figure 8****0 6 . 1**

Calculate the percentage of aluminium used for packaging.

[1 mark]

Percentage of aluminium used for packaging = _____ %

Question 6 continues on the next page**Turn over ►**

0 6 . 2

The formula of aluminium oxide is Al_2O_3

Aluminium oxide contains aluminium ions (Al^{3+}) and oxygen ions (O^{2-}).

Describe what happens when aluminium atoms react with oxygen atoms to produce aluminium oxide.

Refer to electrons in your answer.

[4 marks]

0 6 . 3

The relative formula mass (M_r) of aluminium oxide (Al_2O_3) is 102.

Calculate the percentage by mass of aluminium in aluminium oxide.

Give your answer to **2** significant figures.

Relative atomic mass (A_r): Al = 27

[3 marks]

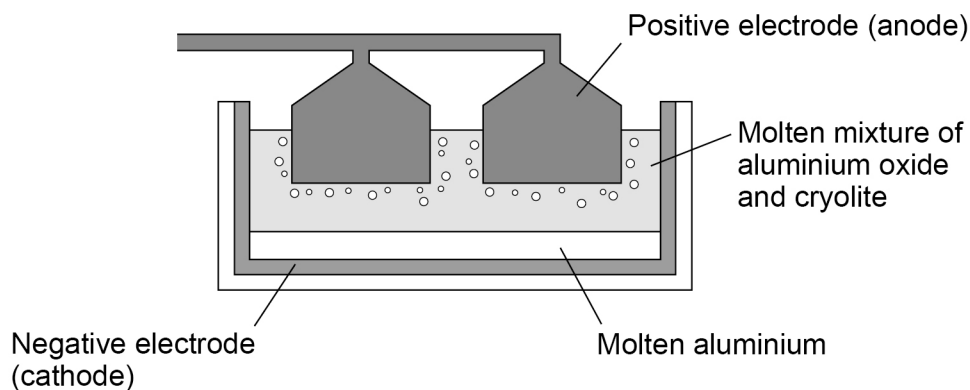
Percentage by mass of aluminium = _____ %



Figure 9 shows the extraction of aluminium from aluminium oxide using electrolysis.

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Figure 9



- 0 6 . 4** The mixture of aluminium oxide and cryolite must be molten so that it conducts electricity.

Why does the mixture conduct electricity when molten?
Refer to particles in your answer.

[1 mark]

- 0 6 . 5** Why is cryolite added to aluminium oxide?

[1 mark]

- 0 6 . 6** Why does aluminium form at the negative electrode rather than the positive electrode?

[1 mark]

Question 6 continues on the next page

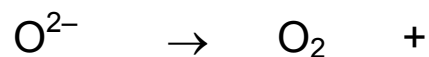
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0 6 . 7 Oxygen is formed at the positive electrode.

Complete and balance the half equation for this reaction.

[2 marks]



0 6 . 8 The positive electrode needs to be replaced at regular intervals.

Explain why.

[2 marks]

15

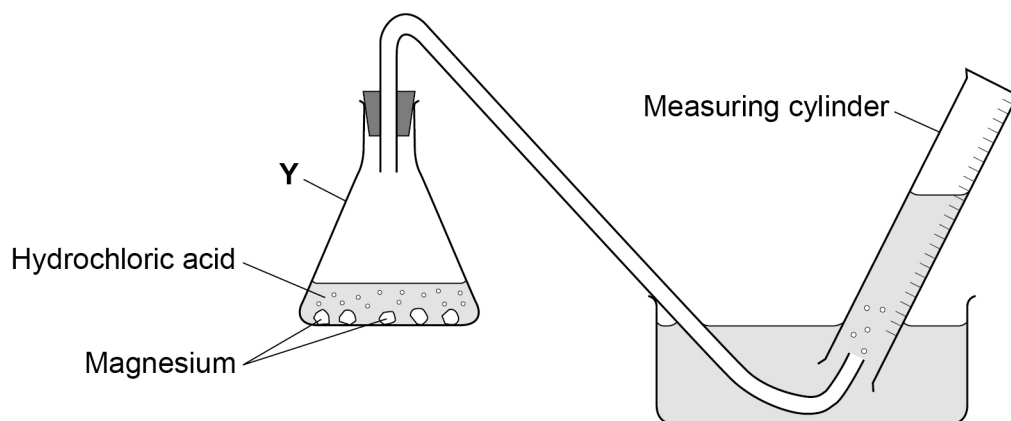


07

A student investigated the rate of the reaction between magnesium and dilute hydrochloric acid.

Figure 10 shows the apparatus used.

Figure 10



07.1

Name the piece of apparatus labelled Y in Figure 10.

[1 mark]

07.2

The student measured the volume of hydrogen gas produced over time.

Figure 10 shows that the student made an error setting up the apparatus.

What problem would this error cause?

[1 mark]

07.3

The gas collected is **not** pure hydrogen.

Why?

[1 mark]

Question 7 continues on the next page

Turn over ►



The student corrected the error shown in **Figure 10**.

Table 8 shows the student's results.

Table 8

Time in seconds	Volume of gas in cm ³
0	0
30	15
60	23
90	26
120	31
150	35
180	38
210	40
240	40



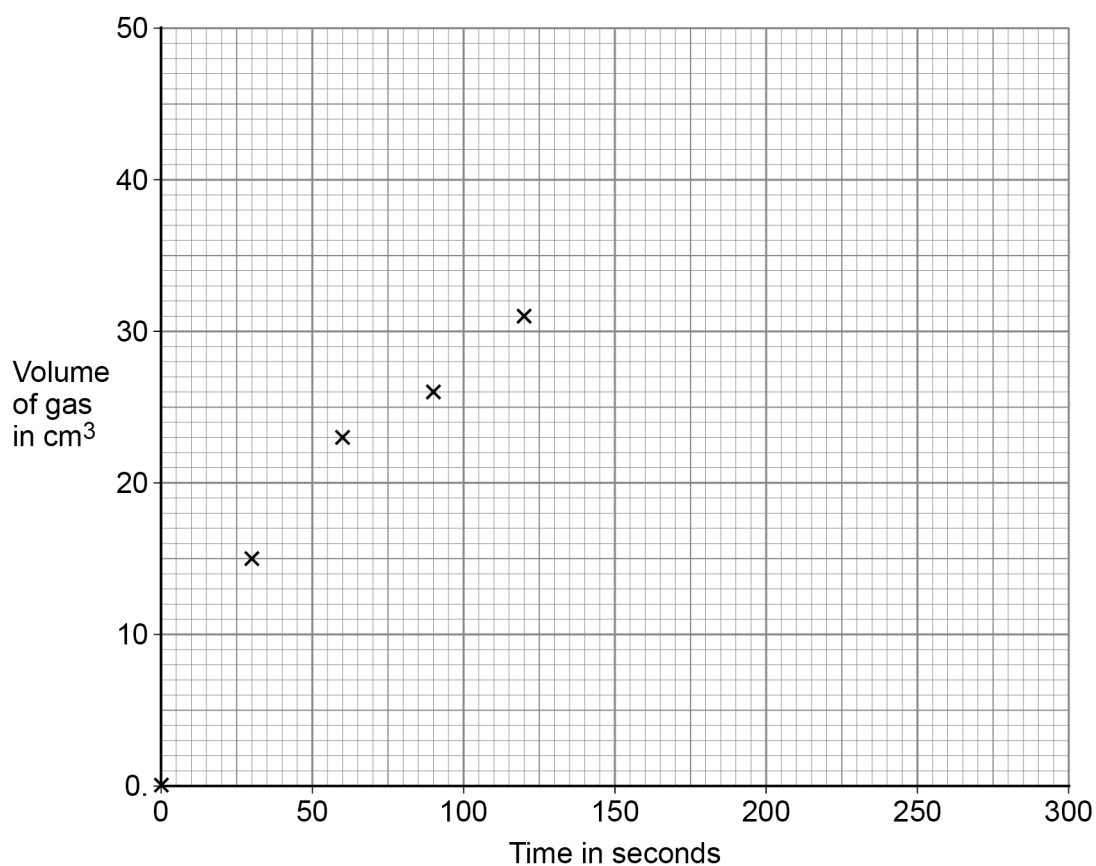
0 7 . 4 Complete the graph in **Figure 11**.

You should:

- plot the remaining data from **Table 8**
- draw a line of best fit.

[3 marks]

Figure 11



0 7 . 5 How does **Figure 11** show that the rate of reaction decreases with time?

[1 mark]

Question 7 continues on the next page

Turn over ►



0 7 . 6

Give **two** reasons why the rate of reaction decreases with time.**[2 marks]**

1 _____

2 _____

0 7 . 7

The student repeated the experiment but included 10 cm³ of water in the reaction mixture.

The quantities of magnesium and hydrochloric acid were not changed.

The magnesium was in excess.

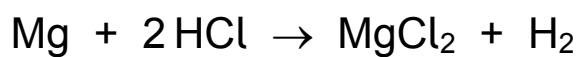
Explain the effect, if any, that the added water would have on the final volume of hydrogen gas produced.

[2 marks]



0 7 . 8

The equation for the reaction is:



The student used:

- 0.0040 mol of magnesium (an excess)
- 17 cm³ of 0.20 mol/dm³ hydrochloric acid.

Calculate by how many moles the magnesium was in excess.

[4 marks]

Number of moles of magnesium in excess = _____

15**END OF QUESTIONS**

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[illegible]

[illegible]

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3 6



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